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Pymble Ladies' College

HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION 2014

Mathematics

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Use pencil for Questions 1-10.
- Write using black or blue pen for Questions 11-16. Black pen is preferred.
- Board approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- In Questions 11-16, show relevant mathematical reasoning and/or calculations.

Total Marks – 100

Section I Pages 1-4

10 marks

- Attempt Questions 1-10
- Allow about 15 mins for this section

Section II Pages 5-15

90 marks

- Attempt Questions 11-16
- Allow about 2 hours 45 minutes for this section

Mark	/100
Highest Mark	/100
Rank	

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section.

Use the multiple choice answer sheet for Questions 1-10.

1 What is the period of $y = 5\sin 2x$?

- (A) π
- (B) 2π
- (C) $\frac{\pi}{2}$
- (D) 5

2 What is the value of x if $8(x-3)^3 - 1 = 0$?

- (A) $3 - \frac{1}{2}$
- (B) $3 + \frac{1}{2}$
- (C) $3 \pm \frac{1}{2}$
- (D) 5

3 What are the values of m that will give the equation $mx^2 + 6x - 3 = 0$ two real and different roots?

- (A) $m \leq -3$
- (B) $m \geq -3$
- (C) $m > -3$
- (D) $m < -3$

Multiple Choice (continued).

4 What is the gradient of the normal to the curve $y = 2x^2 - 5x + 1$ at the point $(2, -1)$?

(A) $\frac{1}{3}$

(B) $-\frac{1}{3}$

(C) 3

(D) -3

5 What is the derivative of $f(x) = \ln(\cos x)$?

(A) $f'(x) = -\tan x$

(B) $f'(x) = \tan x$

(C) $f'(x) = \frac{1}{\cos x}$

(D) $f'(x) = -\frac{1}{\sin x}$

6 What are the solutions of $2\sin x + \sqrt{3} = 0$ in the domain $0 \leq x \leq 2\pi$?

(A) $\frac{\pi}{3}, \frac{2\pi}{3}$

(B) $\frac{2\pi}{3}, \frac{5\pi}{3}$

(C) $\frac{\pi}{3}, \frac{4\pi}{3}$

(D) $\frac{4\pi}{3}, \frac{5\pi}{3}$

Multiple Choice (continued).

7 Given $\log_{10} y = 2 - \log_{10} x$, which expression is equivalent to y ?

(A) $y = \log_{10}(2) - x$

(B) $y = 2 - x$

(C) $y = \frac{100}{x}$

(D) $y = 100 - x$

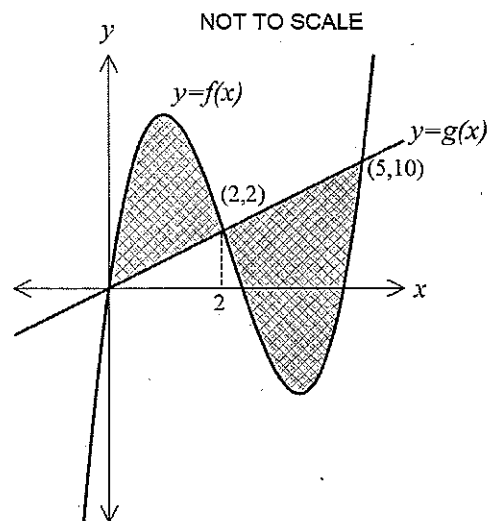
8 Which expression using integral notation is equivalent to the area of the shaded regions?

(A) $\int_0^5 \{f(x) - g(x)\} dx$

(B) $\int_0^2 \{g(x) - f(x)\} dx + \int_2^5 \{f(x) - g(x)\} dx$

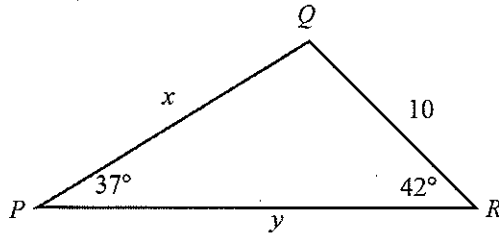
(C) $\int_0^2 \{f(x) - g(x)\} dx + \int_2^5 \{f(x) - g(x)\} dx$

(D) $\int_0^2 \{f(x) - g(x)\} dx + \int_2^5 \{g(x) - f(x)\} dx$



Multiple Choice (continued).

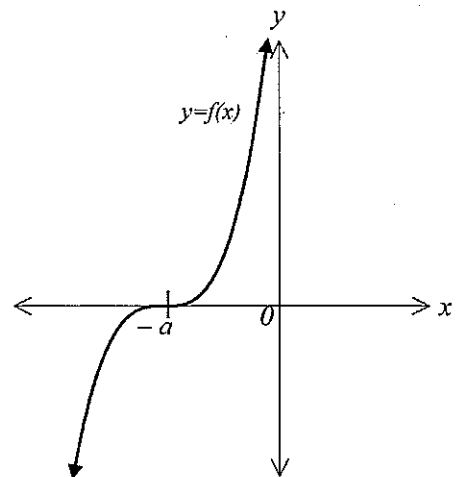
- 9 PQR is a triangle with side lengths x , 10 and y , as shown below.
In this triangle, angle $RPQ = 37^\circ$ and angle $QRP = 42^\circ$.



Which one of the following expressions is correct for triangle PQR ?

- (A) $x = \frac{10}{\sin 37^\circ}$
- (B) $x = 10 \times \frac{\sin 42^\circ}{\sin 37^\circ}$
- (C) $y = 10 \times \frac{\sin 37^\circ}{\sin 101^\circ}$
- (D) $10^2 = x^2 + y^2 - 2xy \cos 42^\circ$
- 10 At $x = -a$, which of the following correctly describes the graph of $y = f(x)$?

- (A) $f(-a) = 0$, $f'(-a) > 0$
- (B) $f'(-a) = 0$, $f''(-a) = 0$
- (C) $f(0) = -a$, $f'(-a) > 0$
- (D) $f'(-a) > 0$, $f''(-a) = 0$



Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section.

Answer each question in the appropriate writing booklet. Extra booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a **separate** writing booklet.

Marks

- (a) Solve the equation $|3 + 5x| = 2$. 2
- (b) Show that $3\sqrt{5} - 2\sqrt{2}$ is a square root of $53 - 12\sqrt{10}$. 2
- (c) Differentiate $(5 - \cos 2x)^4$. 2
- (d) Find a primitive of $\sec^2 x - 3$. 2

Question 11 continues on page 6.

(e) α and β are the roots of $x^2 - 4x + 1 = 0$.

(i) Find $\alpha\beta$.

1

(ii) Hence, prove $\alpha + \frac{1}{\alpha} = 4$.

1

(f) Find the coordinates of the focus of the parabola $x^2 = -32(y - 2)$.

2

(g) A circle is divided into n sectors in such a way that the angles of the sectors are in arithmetic progression. The smallest two angles are 3° and 5° . Find the value of n .

3

End of Question 11

(a) Given that $y = \frac{x^2}{\tan 4x}$, find $\frac{dy}{dx}$. 2

(b) Find $\int \sqrt{7x-2} \, dx$. 2

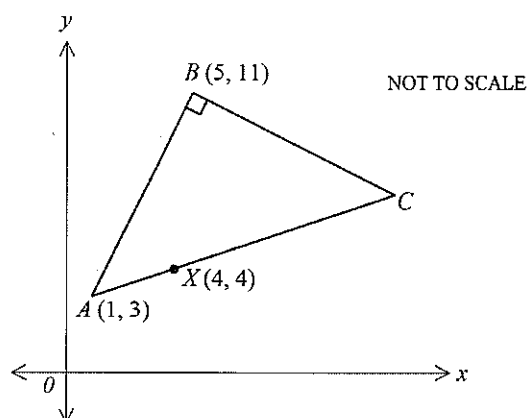
(c) In a geometric series, all the terms are positive, the second term is 24 and the fourth term is $13\frac{1}{2}$. Find

(i) the first term, 2

(ii) the sum to infinity of the series. 1

Question 12 continues on page 8.

(d)



The diagram above shows a triangle ABC in which A has coordinates $(1, 3)$, B has coordinates $(5, 11)$ and angle ABC is 90° . The point $X(4, 4)$ lies on AC . Find

- | | |
|--------------------------------|---|
| (i) the gradient of AB . | 1 |
| (ii) the equation of BC . | 2 |
| (iii) the coordinates of C . | 3 |

- (e) It is given that $f(x) = \frac{1}{x^3} - x^3$. Show that $f(x)$ is a decreasing function. 2

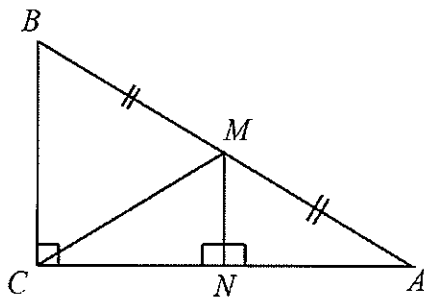
End of Question 12

(a) A function is given by $f(x) = x^3 - 3x^2 - 9x + 11$.

(i) Find the coordinates of the stationary points of $f(x)$ and determine their nature. 3

(ii) Hence, sketch the graph $y = f(x)$ showing all stationary points and the y - intercept. 2

(b) In the diagram, M is the midpoint of AB . $\angle ACB = \angle MNA = 90^\circ$.
Copy the diagram into your booklet.



Prove that

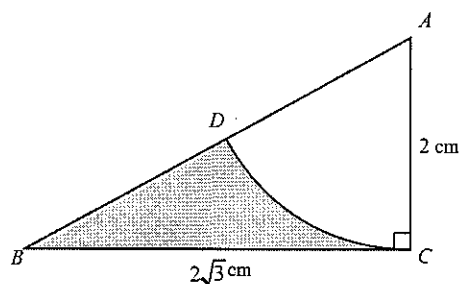
(i) $MN = \frac{1}{2}BC$. 2

(ii) $\triangle AMN$ and $\triangle CMN$ are congruent. 2

(iii) $CM = \frac{1}{2}AB$ 1

Question 13 continues on page 10.

(c)



In the diagram, D lies on the side AB of the triangle ABC and CD is an arc of a circle with centre A and radius 2 cm. The line BC is of length $2\sqrt{3}$ cm and is perpendicular to AC .

2

Find the area of the shaded region BDC , giving your answer in terms of π and $\sqrt{3}$.

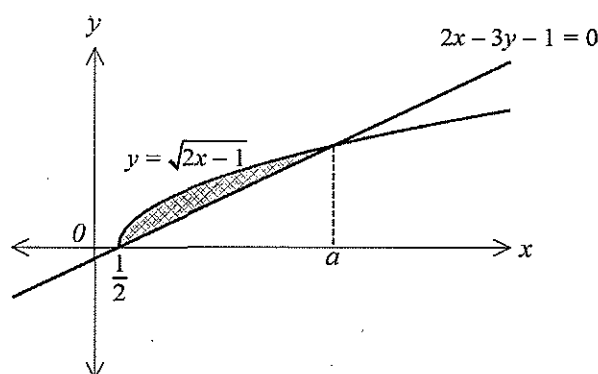
- (d) A curve is such that $\frac{d^2y}{dx^2} = 4e^{-2x}$. Given that $\frac{dy}{dx} = 3$ when $x = 0$ and that the curve passes through the point $(2, e^{-4})$, find the equation of the curve.

3

End of Question 13

(a) Prove that $\frac{\sin A}{1 + \cos A} + \frac{1 + \cos A}{\sin A} = 2 \operatorname{cosec} A$ 3

- (b) The curves $y = \sqrt{2x-1}$ and $2x-3y-1=0$ are drawn below.
They intersect at $x = \frac{1}{2}$ and $x = a$ as indicated on the diagram.



- (i) Show that $a = 5$. 2

- (ii) Find, showing all necessary working, the area of the shaded region. 3

- (c) The temperature $T^{\circ}\text{C}$ of an object in a room, after t minutes, satisfies the differential equation

$$\frac{dT}{dt} = k(T-22), \text{ where } k \text{ is a constant.}$$

- (i) Show that $T = Ae^{kt} + 22$, satisfies the differential equation. 1

- (ii) When $t = 0$, $T = 100$, and when $t = 15$, $T = 70$.

- (α) Use this information to find the values of A and k . 3

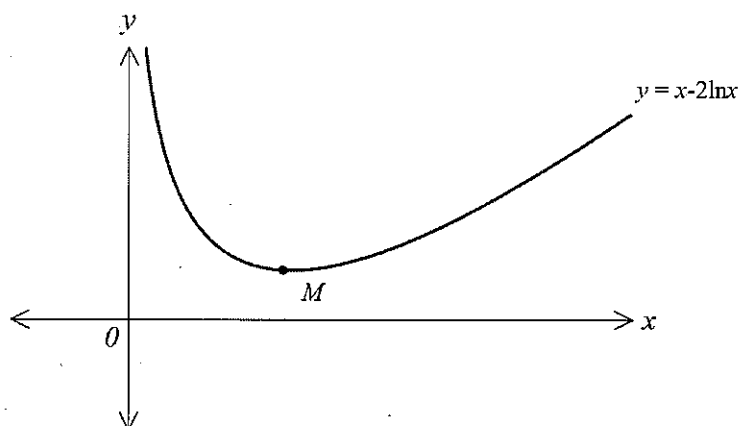
- (β) Hence find the value of t when $T = 40$, correct to 1 decimal place. 3

End of Question 14

(a) Evaluate $\int_2^3 \frac{x^2}{x^3 - 2} dx$.

2

(b)



The above diagram shows the curve $y = x - 2 \ln x$ and its minimum point M .

(i) Find the x coordinate of M .

2

(ii) Use 2 applications of the trapezoidal rule to estimate the value of

3

$$\int_2^4 (x - 2 \ln x) dx.$$

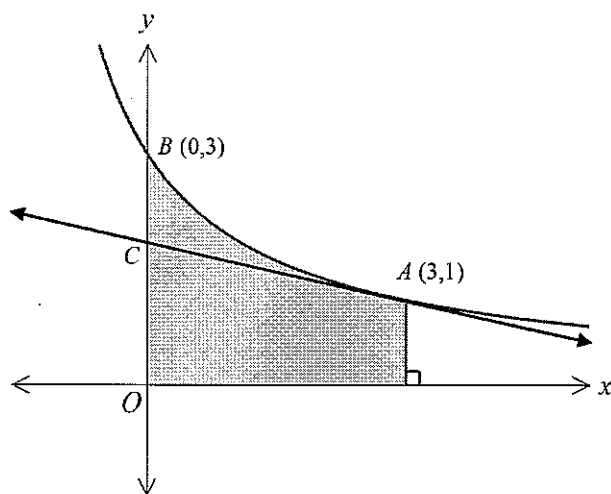
Give your answer to correct to 2 decimal places.

(iii) State, with a reason, whether the trapezoidal rule gives an under-estimate or an over-estimate of the true value of the integral in part (ii).

1

Question 15 continues on page 13.

(c)



The above diagram shows part of the curve $y = \frac{9}{2x+3}$, crossing the y -axis at the point $B(0,3)$.

The point A on the curve has coordinates $(3,1)$.

The tangent to the curve at A crosses the y -axis at C .

- (i) Find the equation of the tangent to the curve at A . 3
- (ii) Determine, showing all necessary working, whether C is nearer to the point B or to the point O . 1
- (iii) Find the exact volume obtained when the shaded region is rotated through 360° about the x -axis. Show all necessary working. 3

End of Question 15

(a) Solve $4e^{2x} - e^x = 0$. **2**

(b) A particle moves in a straight line and at time t it has velocity v , where

$$v = 3t^2 - 2 \sin 3t + 6$$

(i) Find an expression for the acceleration of the particle at time t . **1**

(ii) When $t = \frac{\pi}{3}$, show that the acceleration of the particle is $2\pi + 6$. **1**

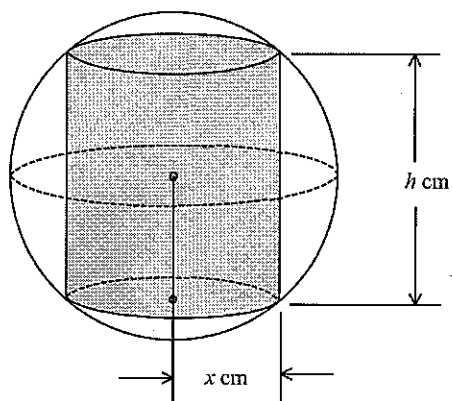
(iii) When $t = 0$, the particle is at the origin.
 Find an expression for the displacement of the particle from the origin at time t . **2**

(c) (i) Find $\frac{d}{dx}(e^{\cos 2x})$. **1**

(ii) Hence, find $\int x + \sin 2x e^{\cos 2x} dx$. **2**

Question 16 continues on page 15.

- (d) A machinist has a spherical ball of brass with diameter 10 cm. The ball is placed in a lathe and machined into a cylinder.



- (i) If the cylinder has radius x cm, show that the cylinder's volume is given by 2
- $$V(x) = \pi x^2 \sqrt{100 - 4x^2} \text{ cm}^3.$$
- (ii) Hence, find the dimensions of the cylinder of largest volume which can be made. 4

End of paper

Mathematics Trial HSC 2017

Solutions

1. A

2. B

3. C

4. B

5. A

6. D

7. C

8. D

9. B

10. B

Question 11

a) $|3 + 5x| = 2$

$$3 + 5x = 2 \quad \text{or} \quad -(3 + 5x) = 2$$

$$5x = -1$$

$$3 + 5x = -2$$

$$x = -\frac{1}{5}$$

or

$$5x = -5$$

(1 mark)

$$x = -1$$

(1 mark)

$$\therefore x = -1 \quad \text{or} \quad -\frac{1}{5}$$

b) $(3\sqrt{5} - 2\sqrt{2})^2 = 45 - 2(2\sqrt{2})(3\sqrt{5}) + 8$ (1 mark)

$$= 53 - 12\sqrt{10}$$

$$\therefore \sqrt{53 - 12\sqrt{10}} = (3\sqrt{5} - 2\sqrt{2})$$

$\therefore 3\sqrt{5} - 2\sqrt{2}$ is a square root of $53 - 12\sqrt{10}$

(1 mark)

c) $\frac{d}{dx} (5 - \cos 2x)^4$

$$= 4(5 - \cos 2x)^3 \cdot 2\sin 2x$$
 (1 mark)

$$= 8\sin 2x (5 - \cos 2x)^3$$
 (1 mark)

d) $\int (\sec^2 x - 3) dx$

$$= \tan x - 3x + C$$

↳ (1 mark)

↘ (1 mark)

e) $x^2 - 4x + 1 = 0$

i) $\alpha\beta = \frac{c}{a}$

$$\alpha\beta = 1$$

(1 mark)

ii) From (i) $\beta = \frac{1}{\alpha}$ and $\alpha + \beta = 4$

Sub $\beta = \frac{1}{\alpha}$ into $\alpha + \beta = 4$

$$\therefore \alpha + \frac{1}{\alpha} = 4$$
 (1 mark)

Question 11 (cont'd)

$$(F) \quad x^2 = -32(y - 2)$$

$$x^2 = -4a(y - 2)$$

$$a = 8$$

(1 mark)

$$\text{Vertex } (0, 2)$$

$$\therefore \text{Focus } (0, -6)$$

(1 mark)

$$g) \quad 3^\circ + 5^\circ + 7^\circ + \dots = 360^\circ \quad (\text{AP})$$

$$a = 3 \quad d = 2 \quad S_n = 360^\circ$$

(1 mark)

$$S_n = \frac{n}{2} (2a + (n-1)d)$$

$$360 = \frac{n}{2} (6 + (n-1)2)$$

$$720 = n(6 + 2n - 2)$$

$$720 = n(4 + 2n)$$

$$\therefore 2n^2 + 4n - 720 = 0$$

(1 mark)

$$n^2 + 2n - 360 = 0$$

$$(n+20)(n-18) = 0$$

$$n = 18 \text{ or } -20$$

$$\text{but } n > 0$$

$$\therefore n = 18$$

(1 mark)

Question 12

a) $y = \frac{x^2}{\tan 4x}$

$$u = x^2$$

$$\frac{du}{dx} = 2x$$

$$v = \tan 4x$$

$$\frac{dv}{dx} = 4 \sec^2 4x$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

(1 mark)

$$\therefore \frac{dy}{dx} = \frac{\tan 4x \cdot 2x - x^2 \cdot 4 \sec^2 4x}{\tan^2 4x}$$

(1 mark)

$$\text{or } \frac{dy}{dx} = \frac{2x \tan 4x - 4x^2 \sec^2 4x}{\tan^2 4x}$$

b) $\int \sqrt{7x-2} \, dx$

$$= \int (7x-2)^{\frac{1}{2}} \, dx$$

$$= \frac{(7x-2)^{\frac{3}{2}}}{\frac{3}{2} \times 7} + C$$

$$= \frac{2 \sqrt{(7x-2)^3}}{21} + C$$

(1 mark)

c) i) $T_2 = 24$
 $T_4 = 13\frac{1}{2}$

GP: $T_n = ar^{n-1}$

$$ar = 24 \quad (1)$$

$$ar^3 = \frac{27}{2} \quad (2)$$

$$(2) \div (1)$$

$$r^2 = \frac{27}{48}$$

$$r = \pm \sqrt{\frac{27}{48}}$$

$$r = \pm \frac{3}{4} \quad (1 \text{ mark})$$

since $r > 0$ then $r = \frac{3}{4}$ and $a = \frac{24}{\frac{3}{4}} = 32 \quad (1) \quad (+)$

Q12 (cont'd)

$$\begin{aligned} \text{c ii)} \quad S_{\infty} &= \frac{a}{1-r} \\ &= \frac{32}{1 - \frac{3}{4}} \end{aligned}$$

NB $|r| < 1$

$$= \frac{32}{\frac{1}{4}}$$

$$= 128$$

(1 mark)

$$\begin{aligned} \text{d i)} \quad M_{AB} &= \frac{11-3}{5-1} \\ &= 2 \end{aligned}$$

$$\text{ii)} \quad M_{BC} = -\frac{1}{2}$$

$$\therefore BC \text{ is } y - 11 = -\frac{1}{2}(x - 5) \quad (1 \text{ mark})$$

$$2y - 22 = -x + 5$$

$$x + 2y - 27 = 0 \quad (1)$$

(1 mark)

$$\text{iii)} \quad M_{AC} = \frac{4-3}{4-1}$$

$$= \frac{1}{3}$$

$$\therefore AC \text{ is } y - 3 = \frac{1}{3}(x - 1)$$

$$3y - 9 = x - 1$$

$$x - 3y + 8 = 0 \quad (2)$$

$$(1) - (2) \quad 5y - 35 = 0$$

$$y = 7$$

(1 mark)

Sub in (1)

$$x = 27 - 2(7)$$

$$x = 13$$

$$\therefore C \text{ is } (13, 7)$$

(1 mark)

(5)

Question 12 (cont'd)

$$e) f(x) = \frac{1}{x^3} - x^3, x \neq 0$$

$$= x^{-3} - x^3$$

For a decreasing function $f'(x) < 0$

$$\begin{aligned} f'(x) &= -3x^{-4} - 3x^2 \\ &= -\frac{3}{x^4} - 3x^2 \end{aligned}$$

For all x , x^2 and $x^4 > 0$

$$\therefore -\frac{3}{x^4} < 0 \text{ and } -3x^2 < 0.$$

$$\therefore f'(x) < 0 \text{ for all } x$$

$\therefore$ decreasing function.

(1 mark for showing correct $f'(x) < 0$
for decreasing function)

(1 mark for qualifying why function
will be < 0 for all x .)

Question 13

a. i) $f(x) = x^3 - 3x^2 - 9x + 11$

$$f'(x) = 3x^2 - 6x - 9$$

For stationary points, $f'(x) = 0$

$$\therefore 3x^2 - 6x - 9 = 0$$

$$3(x^2 - 2x - 3) = 0$$

$$3(x-3)(x+1) = 0$$

$$\therefore x = 3 \text{ or } -1$$

$$(3, -16) \quad (-1, 16) \quad (1 \text{ mark})$$

$$f''(x) = 6x - 6$$

when $x = -1$ $f''(-1) = -6 - 6$

$$= -12 \text{ which is } < 0$$

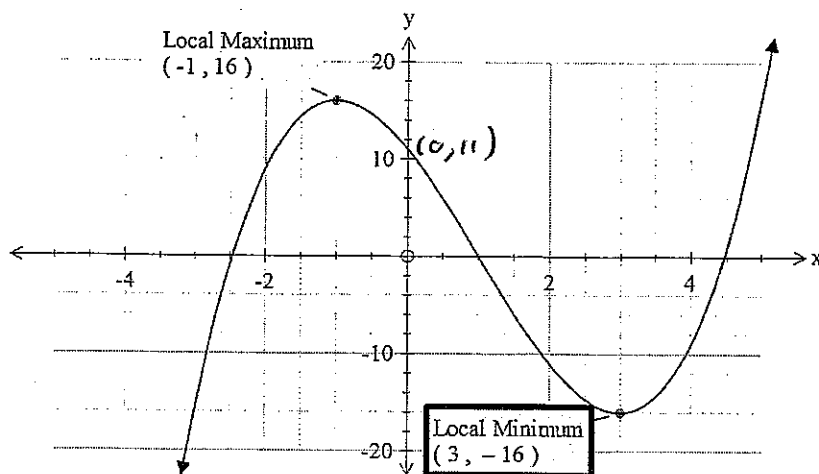
$\therefore$ Maximum turning point at $(-1, 16)$ (1 mark)

when $x = 3$ $f''(3) = 18 - 6$

$$= 12 \text{ which is } > 0$$

$\therefore$ Minimum turning point at $(3, -16)$

(1 mark)



(1 mark for points and scale)

(1 mark for shape)

Question 13 (cont'd)

b i) In $\triangle BCA$ and $\triangle MNA$

$$\angle BCA = \angle MNA = 90^\circ \text{ (given)}$$

$\angle A$ is common

(1 mark)

$\therefore \triangle BCA \equiv \triangle MNA$ (equiangular)

$$\therefore \frac{BA}{MA} = \frac{BA}{\frac{1}{2}BA} = 2 \text{ since } M \text{ is the midpoint of } AB$$

$\therefore$ The reduction factor is $\frac{1}{2}$ (1 mark)

$$\therefore MN = \frac{1}{2}BC \text{ (ratio of corresponding sides in similar triangles)}$$

ii) $\frac{AM}{AB} = \frac{1}{2}$

$$\therefore \frac{AN}{AC} = \frac{1}{2} \text{ (ratio of corresponding sides in similar triangles)}$$

$$AN = \frac{1}{2}AC$$

$$\therefore AN = CN$$

(1 mark)

In $\triangle AMN$ and $\triangle CMN$

$$AN = CN \text{ (shown above)}$$

$$MN = MN \text{ common side}$$

(1 mark)

$$\angle ANM = \angle CNM = 90^\circ \text{ (given)}$$

$$\therefore \triangle AMN \equiv \triangle CMN \text{ (SAS)}$$

iii) $CM = MA$ (corresponding sides in congruent triangles from (i))

$$BM = MA \text{ given}$$

$$\therefore CM = BM$$

(1 mark)

$$CM = \frac{1}{2}BA \text{ (Given } M \text{ is the midpoint of } AB)$$

Question 13 (b) (OR)

$$\begin{aligned} \text{b) in } AB &= AM + BM && \text{(M is the midpoint of AB)} \\ &= AM + AM && \text{(AM = BM)} \\ &= 2AM && \text{--- (1)} \end{aligned}$$

In $\triangle AMN$ and $\triangle ABC$,

$$\sin A = \frac{MN}{AM} = \frac{BC}{AB}$$

$$\Rightarrow MN \cdot AB = BC \cdot AM$$

$$MN \cdot 2AM = BC \cdot AM$$

$$2MN = BC$$

$$MN = \frac{1}{2} BC$$

(2)

ii) $\angle ANM = \angle ACB = 90^\circ$ (given)

Since corresponding angles are equal,

$$MN \parallel BC.$$

$$\frac{AM}{BM} = \frac{AN}{CN} \quad \left(\begin{array}{l} \text{interval parallel to a side of} \\ \text{a triangle divides other sides in} \\ \text{same ratio} \end{array} \right)$$

$$\frac{AN}{CN} = \frac{AM}{AM} \quad (AM = BM)$$

$$= 1$$

$$AN = CN \quad \text{--- (2)}$$

In $\triangle AMN$ and $\triangle CMN$:

$$MN = MN \quad (\text{common})$$

$$\angle ANM = \angle CNM = 90^\circ \quad (\text{given})$$

$$AN = CN \quad (\text{from (2)})$$

$$\therefore \triangle AMN \equiv \triangle CMN \quad (\text{SAS})$$

iii) $CM = AM$ (corresponding sides of congruent triangles)

$$AB = 2AM \quad (\text{from (1)})$$

$$\Rightarrow AM = \frac{1}{2} AB$$

$$CM = \frac{1}{2} AB$$

(8b)

Question 13 (cont'd)

$$\begin{aligned} \text{c) } \tan A &= \frac{2\sqrt{3}}{2} \\ &= \sqrt{3} \\ \therefore A &= \frac{\pi}{3} \end{aligned}$$

$$\begin{aligned} \text{Area BDC} &= \text{Area } \triangle ABC - \text{Area sector ADC} \\ &= \frac{1}{2} \times 2\sqrt{3} \times 2 - \frac{1}{2} \times 2^2 \times \frac{\pi}{3} \quad (1 \text{ mark}) \\ &= 2\sqrt{3} - \frac{2\pi}{3} \text{ cm}^2 \quad (1 \text{ mark}) \end{aligned}$$

$$\text{d) } \frac{d^2y}{dx^2} = 4e^{-2x}$$

$$\frac{dy}{dx} = -\frac{4}{2} e^{-2x} + C \quad (1 \text{ mark})$$

$$\text{When } x=0 \quad \frac{dy}{dx} = 3$$

$$\therefore 3 = -2e^0 + C$$

$$C = 5$$

$$\therefore \frac{dy}{dx} = -2e^{-2x} + 5$$

$$y = e^{-2x} + 5x + k \quad (1 \text{ mark})$$

$$\text{When } x=2 \quad y = e^{-4}$$

$$\therefore e^{-4} = e^{-4} + 5 \times 2 + k$$

$$\therefore k = -10$$

$$\therefore y = e^{-2x} + 5x - 10 \quad (1 \text{ mark})$$

Question 14

$$\begin{aligned} a) \quad LHS &= \frac{\sin A}{1+\cos A} + \frac{1+\cos A}{\sin A} \\ &= \frac{\sin^2 A + (1+\cos A)^2}{(1+\cos A)\sin A} \\ &= \frac{\sin^2 A + 1 + 2\cos A + \cos^2 A}{(1+\cos A)\sin A} \quad (1 \text{ mark}) \\ &= \frac{2 + 2\cos A}{(1+\cos A)\sin A} \quad \text{since } \sin^2 A + \cos^2 A = 1 \\ &= \frac{2(1+\cos A)}{\sin A(1+\cos A)} \quad (1 \text{ mark}) \\ &= \frac{2}{\sin A} \\ &= 2\operatorname{cosec} A \quad (1 \text{ mark}) \\ &= RHS. \\ \therefore LHS &= RHS. \end{aligned}$$

$$\begin{aligned} b) \quad a=5 \quad \text{sub in } y &= \sqrt{2x-1} \\ y &= \sqrt{2 \times 5 - 1} \\ y &= 3 \quad (1 \text{ mark}) \quad a > 0 \end{aligned}$$

$$\begin{aligned} a=5 \quad \text{sub in } 2x-3y-1 &= 0 \quad (1 \text{ for simultaneous equations}) \\ 2 \times 5 - 3y - 1 &= 0 \\ 3y &= 10 - 1 \quad (1 \text{ correct follow}) \\ y &= 3 \quad (1 \text{ mark}) \quad \text{through to} \\ \text{Both equations equal 3 when } a=5 &\quad \text{show } x=5 \\ \therefore \text{2nd pt of intersection is } (5, 3) &\quad \text{as one of 2 solutions} \end{aligned}$$

Question 14 cont'd

$$b) ii) A = \int_{\frac{1}{2}}^5 (2x-1)^{\frac{1}{2}} dx = \frac{1}{2} \times \frac{2}{2} \times 3$$

$$= \left[\frac{2(2x-1)^{\frac{3}{2}}}{3 \times 2} \right]_{\frac{1}{2}}^5 = \frac{27}{4}$$

$$= \frac{9}{3} - 0 = \frac{27}{4}$$

$$= 9 - \frac{27}{4}$$

$$= \frac{9}{4} \text{ sq units.}$$

(1 mark correct integration of $(2x-1)^{\frac{1}{2}}$)

(1 mark correct sub of both expressions)

(1 mark evaluation of substituted value)

$$c) i) \text{ If } T = Ae^{kt} + 22$$

$$\text{or } Ae^{kt} = T - 22$$

$$\frac{dT}{dt} = kAe^{kt}$$

$$\text{but } Ae^{kt} = T - 22$$

(1 mark)

$$\therefore \frac{dT}{dt} = k(T - 22) \text{ as required.}$$

ii) a)

$$100 = Ae^{0k} + 22$$

$$A = 100 - 22$$

$$A = 78$$

(1 mark)

$$\therefore 70 = Ae^{15k} + 22$$

$$70 - 22 = 78e^{15k}$$

$$+8 = 78e^{15k}$$

$$e^{15k} = \frac{48}{78}$$

$$e^{15k} = \frac{8}{13}$$

(1 mark)

$$15k = \ln \frac{8}{13}$$

$$\therefore k = \frac{1}{15} \ln \frac{8}{13} \text{ (1 mark)}$$

(11)

Question 14 (cont'd)

B) When $T \approx 40$ $40 = 78 e^{kt} + 22$

$$e^{kt} = \frac{40 - 22}{78}$$

$$e^{kt} = \frac{3}{13} \quad (1 \text{ mark})$$

$$kt = \ln \frac{3}{13}$$

$$t = \frac{1}{k} \ln \frac{3}{13}$$

$$= \frac{15 \ln \frac{3}{13}}{\ln \frac{8}{13}} \quad (1 \text{ mark})$$

$$= 45.3 \quad (1 \text{ mark})$$

Question 15

$$a) \int_2^3 \frac{x^2}{x^3-2} dx$$

$$= \left[\frac{1}{3} \ln(x^3-2) \right]_2^3 \quad (1 \text{ mark})$$

$$= \frac{1}{3} [\ln 25 - \ln 6] \quad (1 \text{ mark})$$

$$= \frac{1}{3} [2\ln 5 - \ln 6]$$

$$b) i) y = x - 2 \ln x$$

$$\frac{dy}{dx} = 1 - \frac{2}{x} \quad (1 \text{ mark})$$

Stationary points occur at $y' = 0$

$$1 - \frac{2}{x} = 0$$

$$\frac{2}{x} = 1$$

$$x = 2$$

(1 mark)

ii)	x	2	3	4
	$f(x)$	$2-2\ln 2$	$3-2\ln 3$	$4-2\ln 4$

(1 mark)

$$\int_2^4 (x - 2 \ln x) dx \approx \frac{1}{2} [2 - 2\ln 2 + 2(3 - 2\ln 3) + 4 - 2\ln 4] \quad (1 \text{ mark})$$

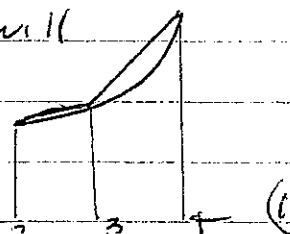
$$= 1.723 \dots$$

$$= 1.72 \text{ to 2 dp.}$$

(1 mark)

iii) Since the curve is concave up for $2 \leq x \leq 4$, the trapezoidal rule will give an over estimate.

(1 mark)



Question 15 (cont'd)

c) $y = \frac{9}{2x+3}$

$$y = 9(2x+3)^{-1}$$

$$\frac{dy}{dx} = -9 \times 2 (2x+3)^{-2} \quad (1 \text{ mark})$$
$$= \frac{-18}{(2x+3)^2}$$

At A when $x=3$

$$\frac{dy}{dx} = \frac{-18}{9^2}$$
$$= -\frac{2}{9} \quad (1 \text{ mark})$$

Tangent at A is: $y-1 = -\frac{2}{9}(x-3)$

$$9y-9 = -2x+6$$

$$2x+9y-15=0 \quad (1 \text{ mark})$$

ii) AC cuts y axis when $x=0$

$$2(0) + 9y - 15 = 0$$

$$9y = 15$$

$$y = \frac{15}{9}$$

$$y = \frac{5}{3}$$

Distance from ~~0~~ O to C is $\frac{5}{3} = 1\frac{2}{3}$

Distance from C to B is $3 - \frac{5}{3} = 1\frac{1}{3}$

$\therefore$ C is closer to B. (1 mark)

Question 15 (cont'd)

$$c. iii) \quad V = \pi \int_0^3 y^2 dx$$

$$= \pi \int_0^3 \left[\frac{9}{(2x+3)} \right]^2 dx \quad (1 \text{ mark})$$

$$= 81\pi \int_0^3 (2x+3)^{-2} dx$$

$$= 81\pi \left[\frac{-1}{2(2x+3)} \right]_0^3 \quad (1 \text{ mark})$$

$$= -\frac{81\pi}{2} \left(\frac{1}{6+3} - \frac{1}{3} \right)$$

$$= -\frac{81\pi}{2} \left(\frac{1}{9} - \frac{1}{3} \right)$$

$$= 9\pi \text{ cubic units.} \quad (1 \text{ mark})$$

Question 16

$$a) \quad 4e^{2x} - e^x = 0$$

$$\therefore 4(e^x)^2 - e^x = 0$$

$$\text{Let } m = e^x$$

$$4m^2 - m = 0$$

$$m(4m - 1) = 0$$

$$\therefore 4m - 1 = 0$$

$$m = \frac{1}{4}$$

$$\therefore e^x = \frac{1}{4}$$

$$\ln e^x = \ln \frac{1}{4}$$

$$x = \ln \frac{1}{4} \quad (1 \text{ mark})$$

$$x = \ln 2^{-2}$$

$$x = -2 \ln 2$$

$$\text{or } m = 0$$

no soln
(as $e^x > 0$)
(1 mark)

$$b) \quad i) \quad v = 3t^2 - 2\sin 3t + 6$$

$$a = 6t - 6\cos 3t \quad (1 \text{ mark})$$

$$ii) \quad \text{When } t = \frac{\pi}{3} \quad a = 6\left(\frac{\pi}{3}\right) - 6\cos\left(3 \times \frac{\pi}{3}\right)$$

$$= 2\pi - 6\cos \pi$$

$$= 2\pi - 6(-1) \quad (1 \text{ mark})$$

$$= 2\pi + 6$$

$$iii) \quad \text{Displacement } x = t^3 + \frac{2}{3}\cos 3t + 6t \quad (1 \text{ mark})$$

$$\text{When } t = 0 \quad x = 0$$

$$0 = 0 + \frac{2}{3} \times 1 + 0 + c$$

$$c = -\frac{2}{3}$$

$$\therefore x = t^3 + \frac{2}{3}\cos 3t + 6t - \frac{2}{3} \quad (1 \text{ mark})$$

Question 16 (cont'd)

c) i) $\frac{d}{dx} e^{\cos 2x} = -2 \sin 2x e^{\cos 2x}$ (1 mark)

ii) $\int x + \sin 2x e^{\cos 2x} dx$

$$= \frac{x^2}{2} - \frac{1}{2} \int -2 \sin 2x e^{\cos 2x} dx$$

$$= \frac{x^2}{2} - \frac{1}{2} e^{\cos 2x} + C \quad (2 \text{ marks})$$

(1 mark for partial x²)

d) i) $V = A \times H$

$$= \pi r^2 \times H$$

$$= \pi \times x^2 \times h$$

Now $x^2 + \left(\frac{h}{2}\right)^2 = 25$ (1 mark)

$$\therefore \frac{h^2}{4} = 25 - x^2$$

$$h^2 = 100 - 4x^2$$

$$h = \sqrt{100 - 4x^2} \quad h > 0 \quad (1 \text{ mark})$$

$$\therefore V(x) = \pi x^2 \sqrt{100 - 4x^2} \text{ cm}^3$$

ii) $V'(x) = 2\pi x \sqrt{100 - 4x^2} + \pi x^2 \cdot \frac{1}{2} (100 - 4x^2)^{-\frac{1}{2}} \cdot -8x$

$$= 2\pi x \sqrt{100 - 4x^2} - \frac{4\pi x^3}{\sqrt{100 - 4x^2}} \quad (1 \text{ mark})$$

$$= \frac{2\pi x (100 - 4x^2) - 4\pi x^3}{\sqrt{100 - 4x^2}}$$

For maximum volume $V'(x) = 0$

Question 16 (cont'd)

d)

$$\therefore 2\pi x(100 - 4x^2 - 2x^2) = 0$$

$$\therefore x = 0 \quad \text{or} \quad 100 - 6x^2 = 0$$

$$x^2 = \frac{100}{6}$$

$$\therefore x = \pm \frac{5\sqrt{2}}{\sqrt{3}}$$

$$x = \pm \frac{5\sqrt{6}}{3} \quad (1 \text{ mark})$$

$$\text{but } x > 0 \quad \therefore x = \frac{5\sqrt{6}}{3}$$

Check if $x = \frac{5\sqrt{6}}{3}$ gives maximum volume

x	0	1	$\frac{5\sqrt{6}}{3}$	4.5
$V'(x)$	0	$\frac{188\pi}{\sqrt{96}}$	0	$\frac{171\pi - 364.5\pi}{\sqrt{9}}$
$V''(x)$	0	> 0	0	< 0

(1 mark)

$\therefore x = \frac{5\sqrt{6}}{3}$ gives a maximum value.

Dimensions Radius $\frac{5\sqrt{6}}{3}$ cm

$$\text{Height } \sqrt{100 - 4\left(\frac{5\sqrt{6}}{3}\right)^2}$$

$$= \sqrt{\frac{100}{3}}$$

$$= \frac{10}{\sqrt{3}} \quad (1 \text{ mark})$$

$$\therefore h = \frac{10\sqrt{3}}{3} \text{ cm}$$